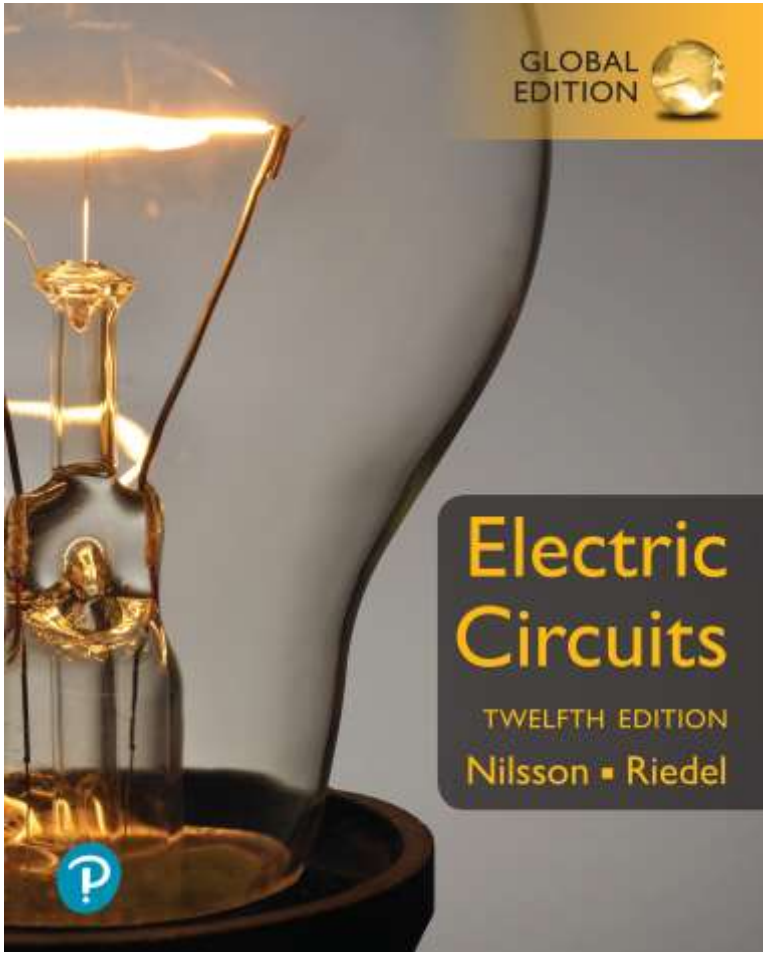
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Twelfth Edition, Global Edition



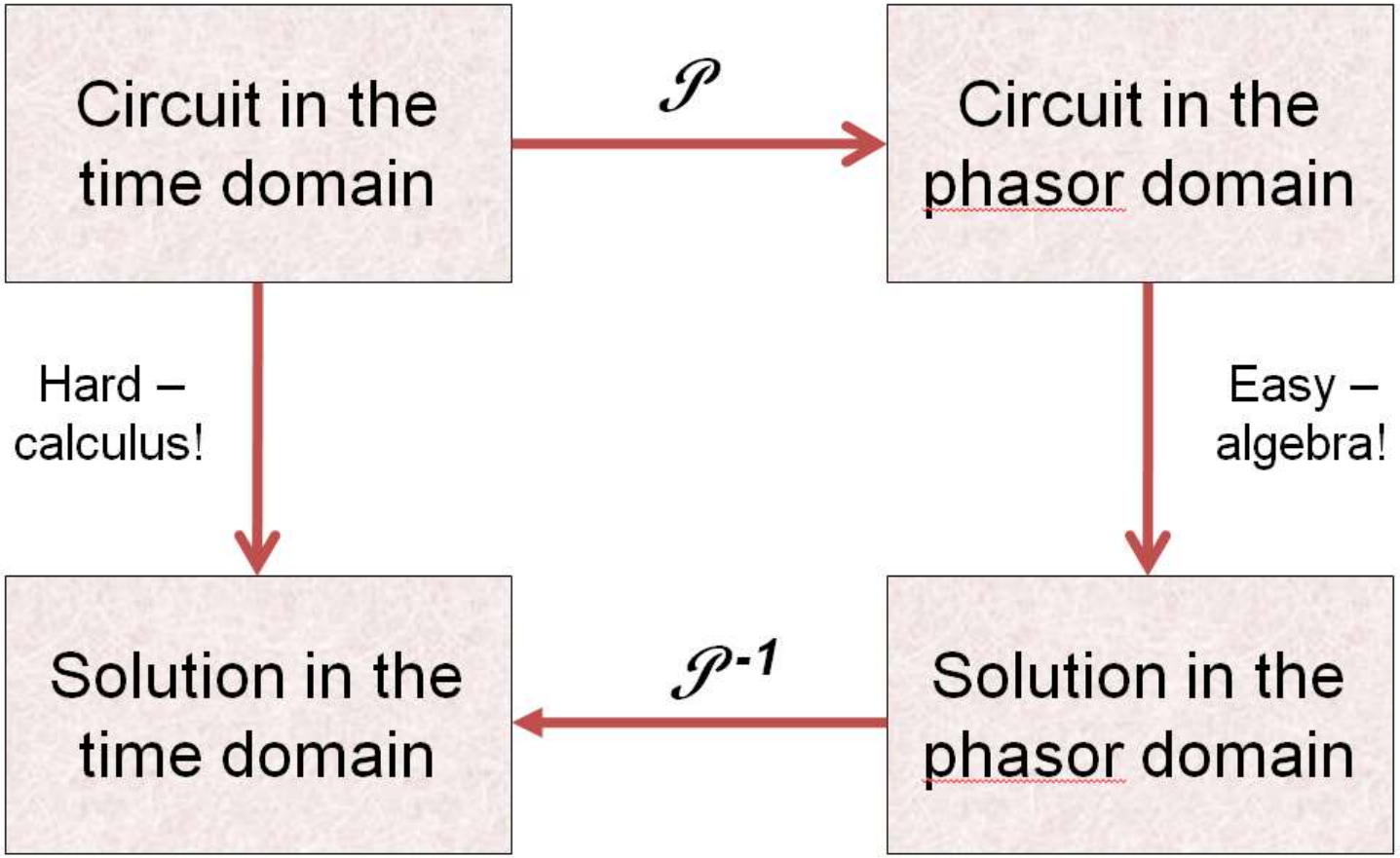
Chapter 9 – Part 2

Sinusoidal Steady-State Analysis

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Chapter 9 – Sinusoidal Steady-State Analysis (1 of 17)



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Chapter 9 – Sinusoidal Steady-State Analysis (2 of 17)

Steps in ACSS Analysis:

1. Redraw the circuit (the phasor transform does not change the components or their connections).
2. Phasor transform all known v(t) and i(t).
3. Represent unknown voltages and currents with *V* and *I*.
4. Replace component values with impedance (Z) values.
5. Use any circuit analysis method(s) to write equations and solve them with a calculator.
6. Inverse-transform the result, which is a phasor, back to the time domain.

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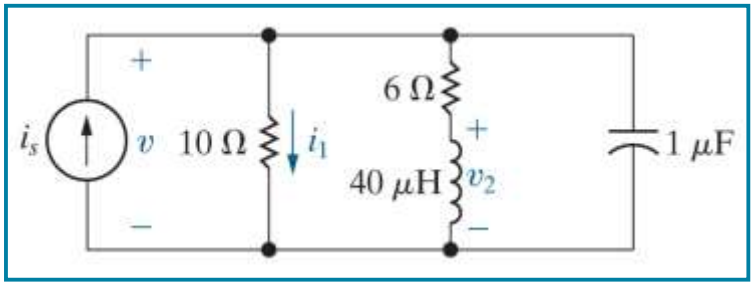
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Chapter 9 – Sinusoidal Steady-State Analysis (3 of 17)

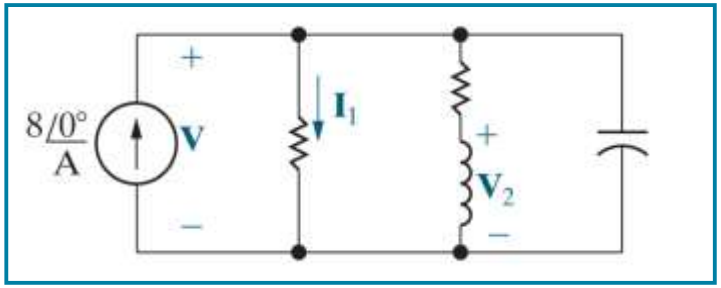
Example:

$i_s(t) = 8\cos(200{,}000t)$ A.

Find $v(t)$, $i_1(t)$, and $v_2(t)$ in the steady-state.



1. Redraw the circuit.
2. Phasor transform all known voltages and currents.
3. Represent unknown voltages and currents with phasor symbols



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Chapter 9 – Sinusoidal Steady-State Analysis (4 of 17)

The next step is to calculate the impedance of all resistors, inductors, and capacitors. Actually, the impedance of the resistors doesn't require calculation.

- ✔ A. True
- ✘ B. False

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Chapter 9 – Sinusoidal Steady-State Analysis (5 of 17)

The impedance of inductors and capacitors is calculated using

- ✗ A. The source frequency
- ✗ B. The component value
- ✓ C. Both A and B

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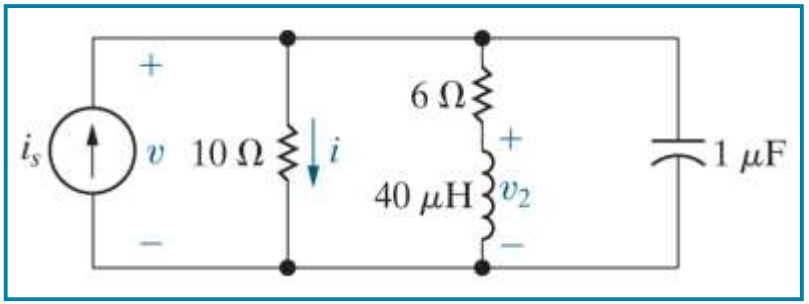
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Chapter 9 – Sinusoidal Steady-State Analysis (6 of 17)

Example (continued):

$i_s(t) = 8\cos(200{,}000t)$ A.

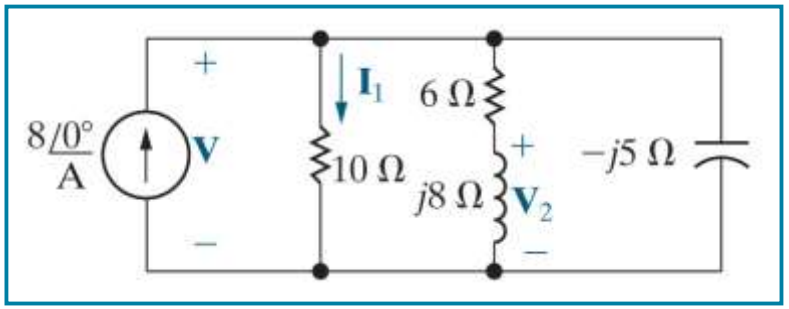
Find $v(t)$, $i_1(t)$, and $v_2(t)$ in the steady-state.



4. Calculate the impedances of all resistors, inductors, and capacitors.

$$Z_L = j\omega L = j(200{,}000)(40\mu) = j8\Omega$$

$$Z_C = \frac{-j}{\omega C} = \frac{-j}{(200{,}000)(1\mu)} = -j5\Omega$$



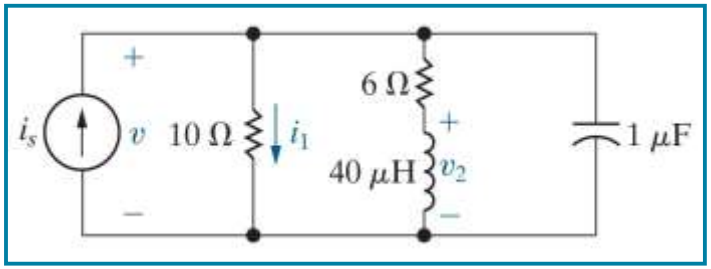
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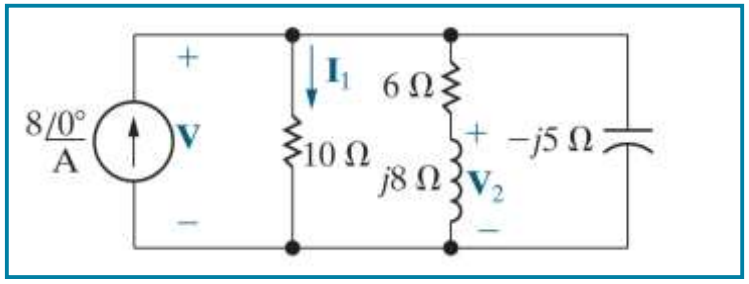
Chapter 9 – Sinusoidal Steady-State Analysis (7 of 17)

Example (continued):

$i_s(t) = 8\cos(200{,}000t)$ A. Find $v(t)$, $i_1(t)$, and $v_2(t)$ in the steady-state.



5. Use DC circuit analysis techniques to find the phasor $\boldsymbol{V}$.
   - In this problem, we'll simplify the circuit by replacing all impedances by one equivalent impedance, and use Ohm's law for phasors.



$$Z_{eq} = 10 \,||\, (6 + j8) \,||\, -j5 = (4 - j3)\Omega$$

$$\boldsymbol{V} = Z\boldsymbol{I} = (4 - j3)(8\angle 0°) = (32 - j25)\text{V}$$

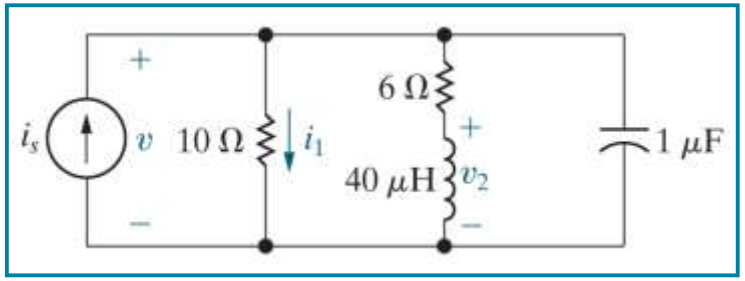
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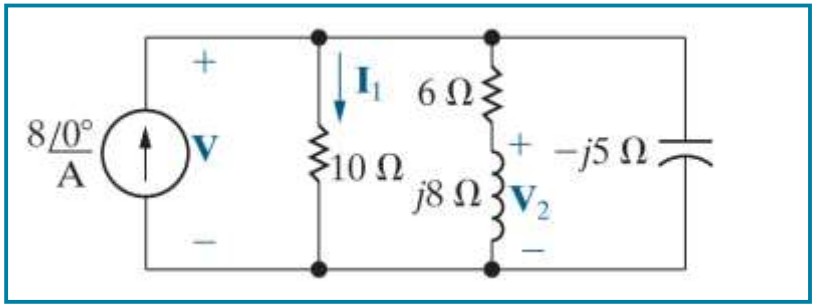
Chapter 9 – Sinusoidal Steady-State Analysis (8 of 17)

Example (continued):

$i_s(t) = 8\cos(200{,}000t)$ A. Find $v(t)$, $i_1(t)$, and $v_2(t)$ in the steady-state.



6. Inverse phasor-transform the result to get back to the time domain.



$$\boldsymbol{V} = (32 - j24)\text{V} = 40\angle -36.87^\circ \text{ V}$$

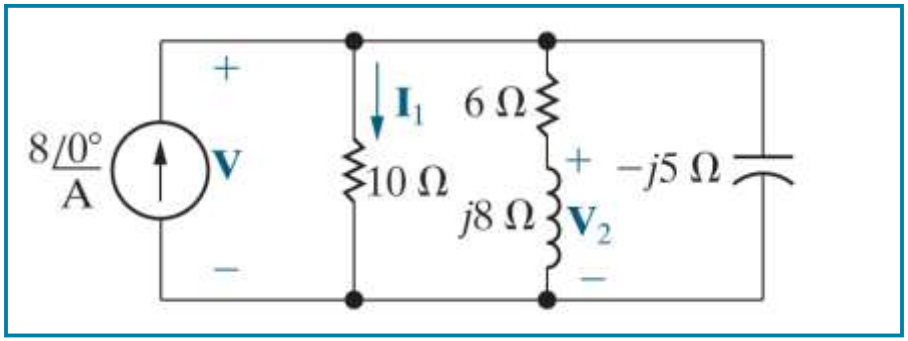
$$v(t) = \mathcal{P}^{-1}\{40\angle -36.87^\circ\} = 40\cos(200{,}000t - 36.87^\circ) \text{ V}$$

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Chapter 9 – Sinusoidal Steady-State Analysis (9 of 17)

The most direct way to calculate the phasor I_1 is



✗ A. Voltage division

✓ B. Current division

✗ C. Source transform

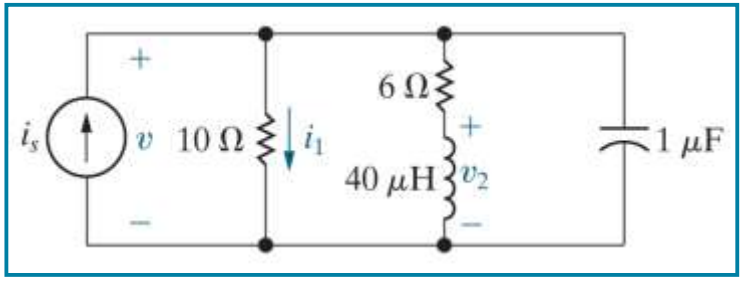
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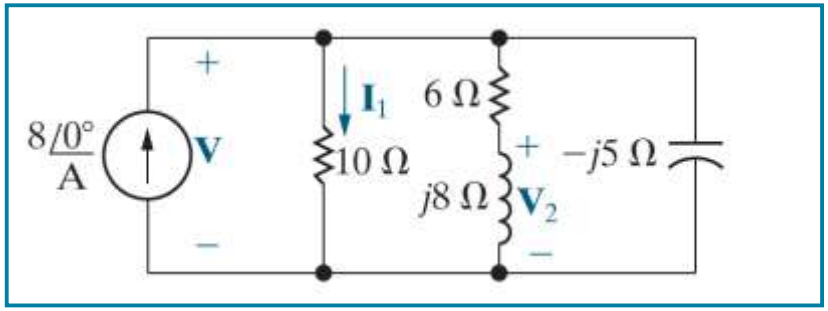
Sinusoid Steady-State Analysis

Example (continued):

$i_s(t) = 8\cos(200{,}000t)$ A. Find $v(t)$, $i_1(t)$, and $v_2(t)$ in the steady-state.



5. Use DC circuit analysis techniques to find the phasor $\boldsymbol{I}_1$.
   - Use current division.



$$\boldsymbol{I}_1 = \frac{Z_{eq}}{Z_1}\boldsymbol{I}_s = \frac{(4 - j3)}{10}(8\angle 0°)$$

$$= (3.2 - j2.4)\ A$$

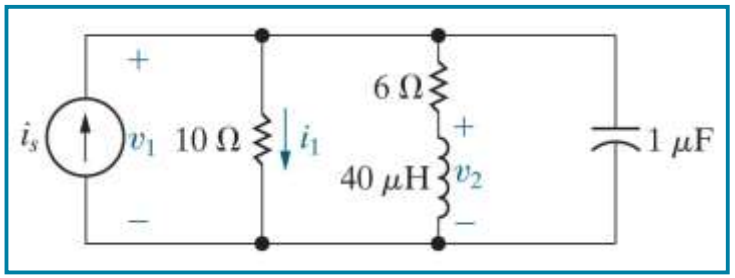
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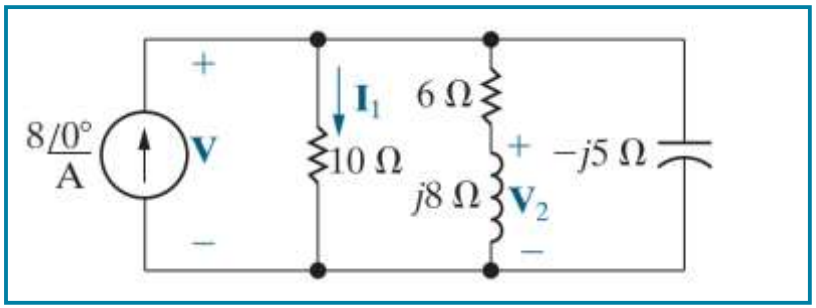
Chapter 9 – Sinusoidal Steady-State Analysis (10 of 17)

Example (continued):

$i_s(t) = 8\cos(200{,}000t)$ A. Find $v(t)$, $i_1(t)$, and $v_2(t)$ in the steady-state.



6. Inverse phasor-transform the result to get back to the time domain.



$$\boldsymbol{I}_1 = (3.2 - j2.4)\ A = 4\angle -36.87^\circ\ A$$

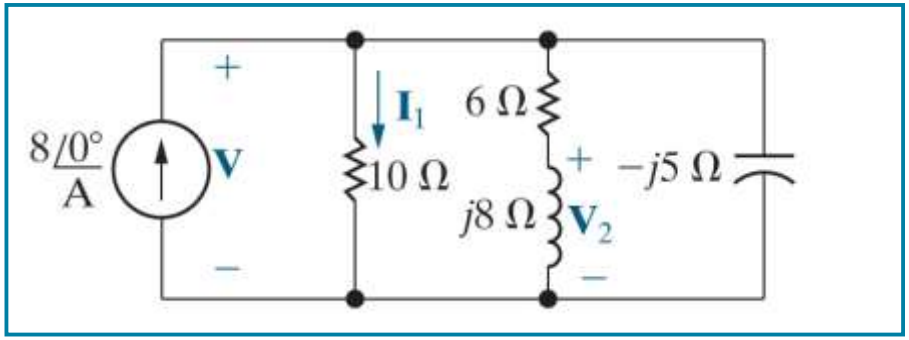
$$i_1(t) = \mathcal{P}^{-1}\{4\angle -36.87^\circ\} = 4\cos(200{,}000t - 36.87^\circ)\ A$$

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Chapter 9 – Sinusoidal Steady-State Analysis (11 of 17)

The most direct way to calculate the phasor V_2 is



- ✔ A. Voltage division
- ✘ B. Current division
- ✘ C. Node voltage method

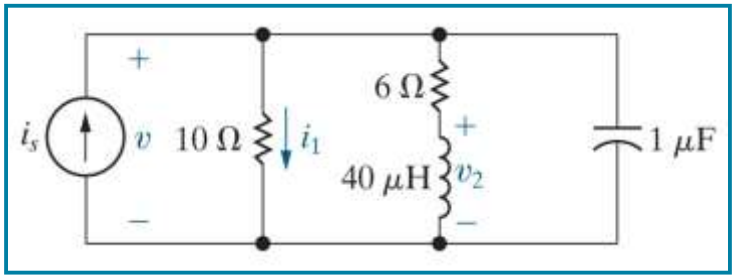
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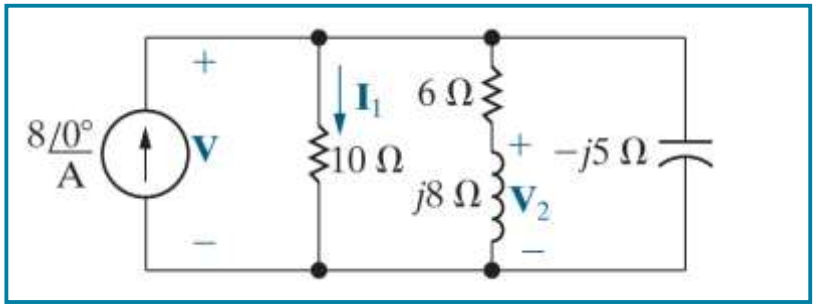
Chapter 9 – Sinusoidal Steady-State Analysis (12 of 17)

Example (continued):

$i_s(t) = 8\cos(200{,}000t)$ A. Find $v(t)$, $i_1(t)$, and $v_2(t)$ in the steady-state.



5. Use DC circuit analysis techniques to find the phasor $\boldsymbol{V}_2$.
   - Use voltage division.



$$\boldsymbol{V}_2 = \frac{Z_L}{Z_R + Z_L}\boldsymbol{V} = \frac{j8}{6 + j8}(32 - j24)$$
$$= (32 + j0)\ V$$

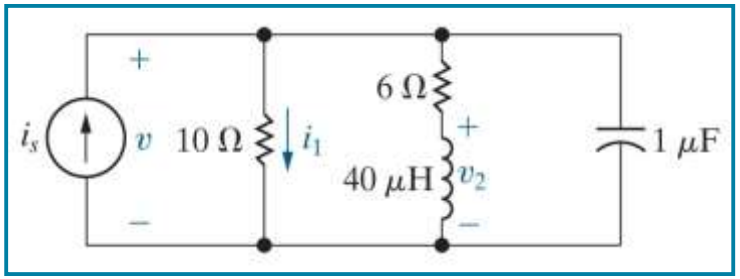
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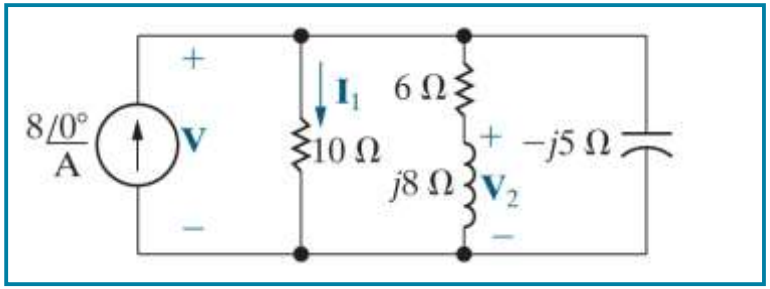
Chapter 9 – Sinusoidal Steady-State Analysis (13 of 17)

Example (continued):

$i_s(t) = 8\cos(200{,}000t)$ A. Find $v(t)$, $i_1(t)$, and $v_2(t)$ in the steady-state.



6. Inverse phasor-transform the result to get back to the time domain.



$$\boldsymbol{V}_2 = (32 + j0)\ V = 32\angle 0°\ V$$

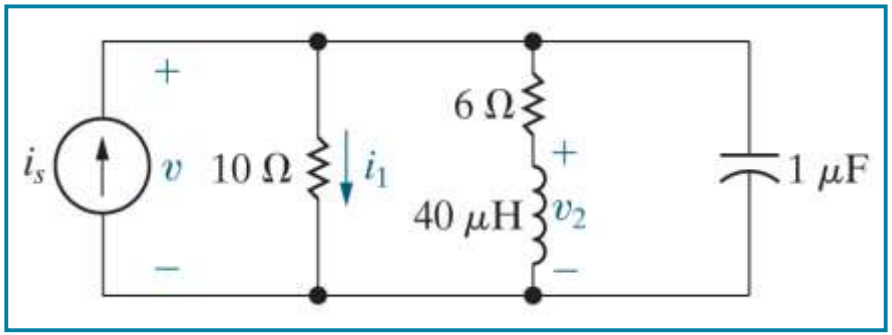
$$v_2(t) = \mathcal{P}^{-1}\{32\angle 0°\} = 32\cos 200{,}000t\ V$$

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Chapter 9 – Sinusoidal Steady-State Analysis (14 of 17)

Suppose we can vary the frequency of the current source. What frequency will cause $i_s(t)$ and *v(t)* to be <u>in-phase</u> in the steady-state?

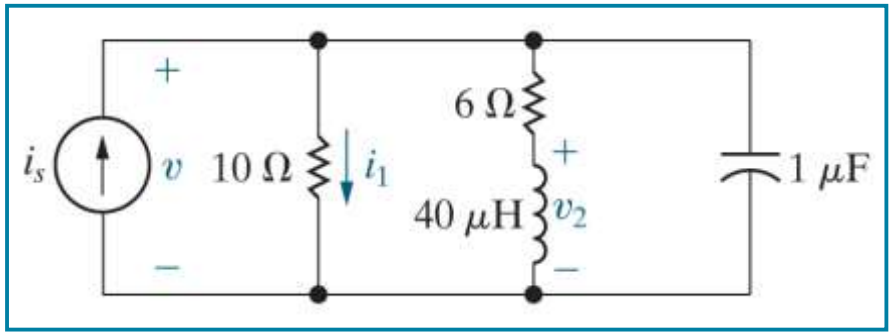


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Chapter 9 – Sinusoidal Steady-State Analysis (15 of 17)

If $i_s(t)$ and *v(t)* are in-phase in the steady-state, this means

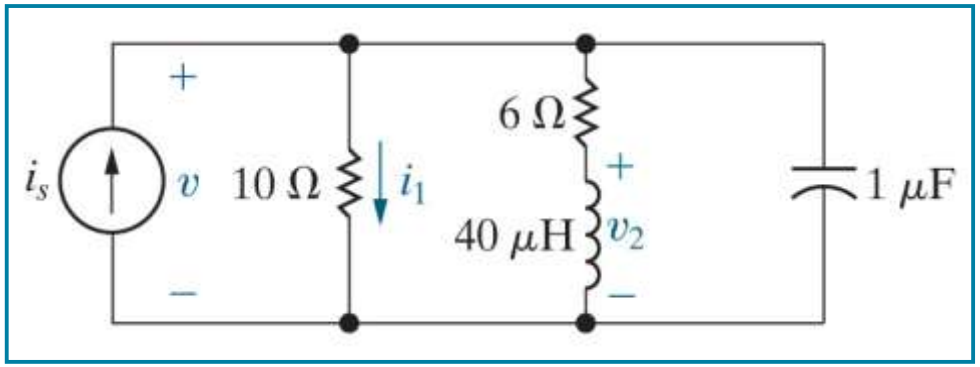


- ✗ A. Their phase angles are the same.
- ✗ B. The equivalent impedance seen by the current source has a phase angle of 0°.
- ✗ C. The equivalent impedance seen by the current source is purely resistive.
- ✓ D. All of the above.

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Chapter 9 – Sinusoidal Steady-State Analysis (16 of 17)



$$\frac{1}{Z_{eq}} = \frac{1}{10} + \frac{1}{6 + j\omega L} + j\omega C$$

$$= \frac{(6 + j\omega L)(6 - j\omega L)}{10(6 + j\omega L)(6 - j\omega L)} + \frac{10(6 - j\omega L)}{10(6 + j\omega L)(6 - j\omega L)} + \frac{10(j\omega C)(6 + j\omega L)(6 - j\omega L)}{10(6 + j\omega L)(6 - j\omega L)}$$

$$= \frac{36 + \omega^2 L^2 + 60 - 10 j\omega L + 10 j\omega C(36 + \omega^2 L^2)}{10(36 + \omega^2 L^2)}$$

$$\mathcal{I}m\left\{\frac{1}{Z_{eq}}\right\} = \frac{-10\omega L + 10\omega C(36 + \omega^2 L^2)}{10(36 + \omega^2 L^2)} = 0 \qquad \therefore \quad -10\omega L + 10\omega C(36 + \omega^2 L^2) = 0$$

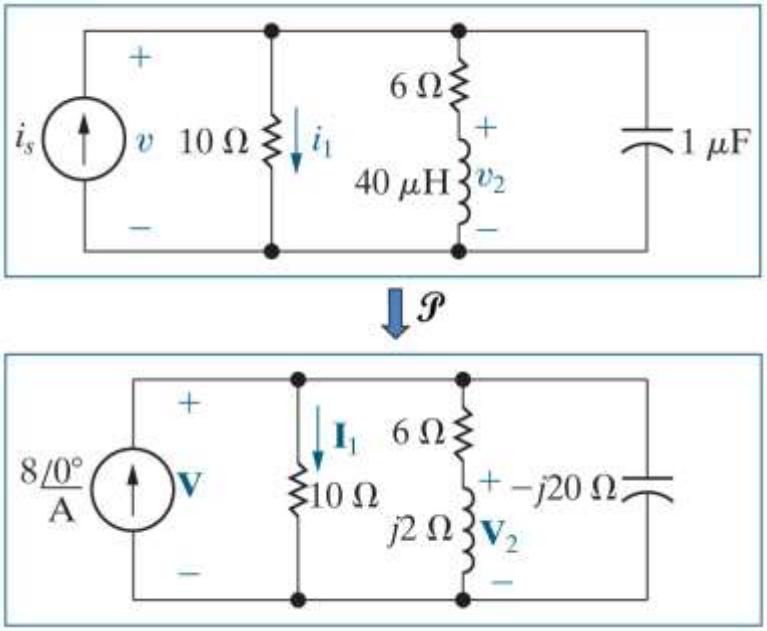
$$\therefore \quad \omega^2 = \frac{L/C - 36}{L^2} = \frac{40/1 - 36}{(40\mu)^2} \qquad \therefore \quad \omega = 50{,}000 \text{ rad/s}$$

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Chapter 9 – Sinusoidal Steady-State Analysis (17 of 17)

Suppose we can vary the frequency of the current source. What frequency will cause $i_s(t)$ and *v(t)* to be in-phase in the steady-state?



For $\omega = 50{,}000$ rad/s

$$Z_L = j\omega L = j(50{,}000)(40\mu) = j2\Omega$$

$$Z_C = \frac{-j}{\omega C} = \frac{-j}{(50{,}000)(1\mu)} = -j20\Omega$$

$$\therefore \quad Z_{eq} = 10 \,||\, (6 + j2) \,||\, -j20 = 4\Omega$$

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